INCH-POUND

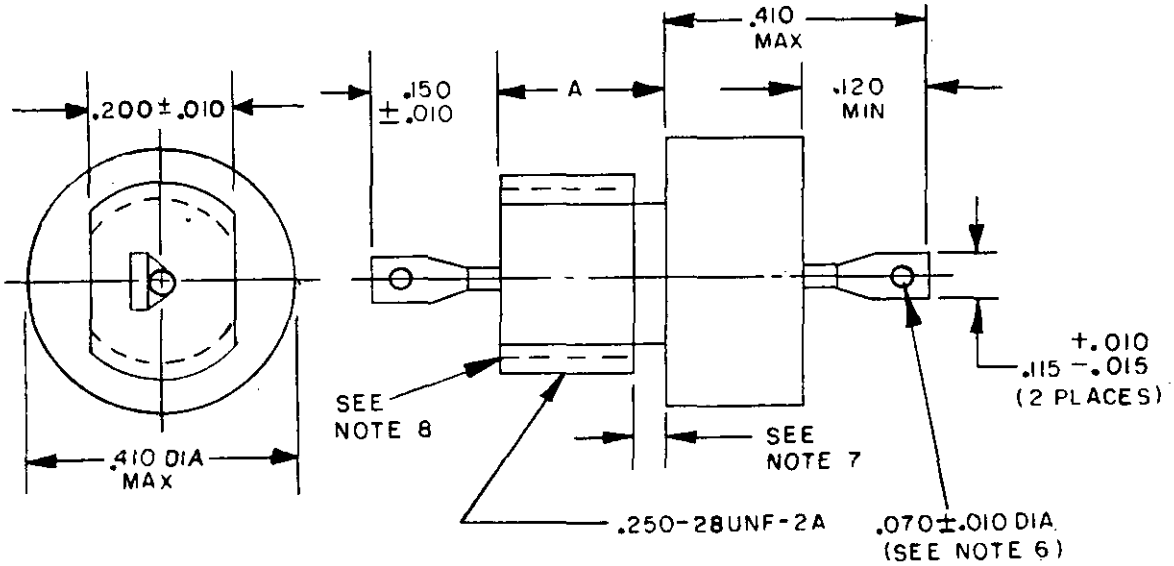
MIL-F-28861/1A
2 November 1988
SUPERSEDING
MIL-F-28861/1
17 December 1981

MILITARY SPECIFICATION SHEET

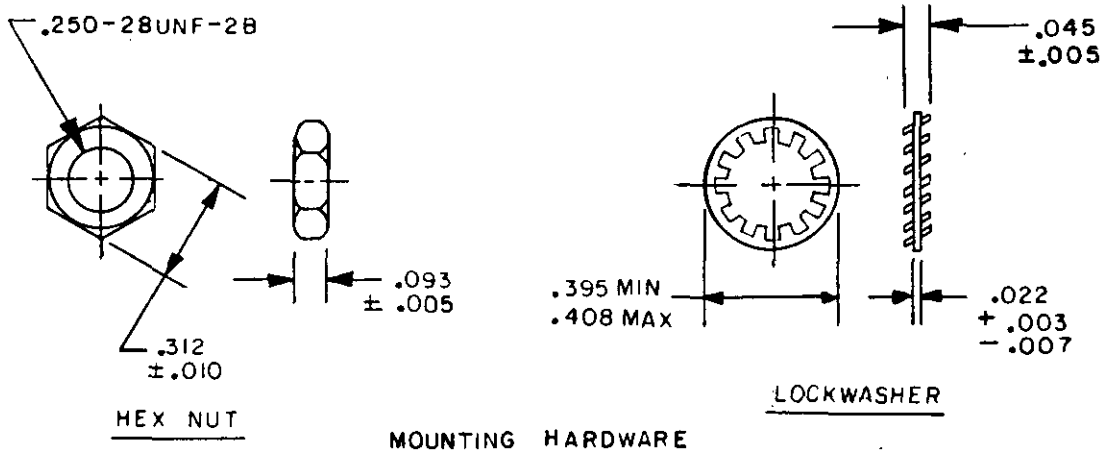
FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLES FS10, FS11, FS12, FS13, AND FS14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.



Styles FS10, FS11, FS12, FS13, and FS14



MOUNTING HARDWARE

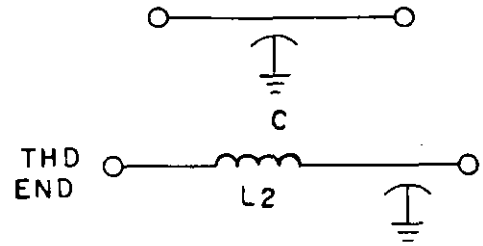
FIGURE 1. Case and hardware dimensions and circuit diagrams.

(A) Denotes changes

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Dash number	A dimension (±.010)
001 thru 010	.187
011 thru 020	.312



CIRCUIT DIAGRAMS

Inches	mm	Inches	mm
.003	0.08	.115	2.92
.005	0.13	.120	3.05
.007	0.18	.150	3.81
.010	0.25	.187	4.75
.015	0.38	.200	5.08
.022	0.56	.250	6.35
.045	1.14	.312	7.92
.070	1.78	.395	10.03
.093	2.36	.408	10.36
		.410	10.41

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagrams are for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. Terminal identification (non-symmetrical filters): The case shall be marked at the threaded end of the filter with the symbol "L" or the circuit diagram shall be marked on the case.
6. Optional slot may be supplied, .050 ±.010 inches (1.27 ±0.25 mm) x .070 ±.010 inches (1.78 ±0.25 mm).
7. Imperfect thread or undercut optional .050 inch (1.27 mm) maximum.
8. One imperfect thread allowed .035 inch (0.89 mm) maximum.
9. Recommended mounting torque: 44 oz-in ±4 oz-in.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 5 grams maximum.

Case finish:

Class B filters, T only (tin plated or tin-lead plated).

(A) Classes A and S filters, T or G only (tin plated, tin-lead plated, or gold plated). In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: As specified in table I.

Rated current: 15 amperes.

Capacitance: See table I for minimum capacitance value.

Dissipation factor: 3 percent maximum.

Voltage and temperature limits of capacitance: +15 percent, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohms minimum or 1,000 megohm-microfarads minimum, whichever is less.

At +125°C: 100 megohms minimum or 100 megohm-microfarads minimum, whichever is less.

Insertion loss: See table I.

Voltage drop: 0.12 V dc maximum.

DC resistance: 0.008 ohm maximum.

(A) Seal: In accordance with MIL-F-28861. Leakage rate for classes A and S filters shall not exceed 1×10^{-7} atm cm^3/s.

Temperature rise: +25°C maximum.

Solderability of terminals: In accordance with MIL-F-28861.

(A) Quality assurance provisions:

Class B and class S filters: In accordance with MIL-F-28861.

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Class A filters:

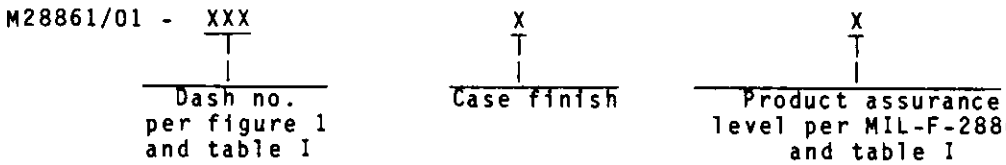
Qualification inspection: Initial qualification may be granted based on qualification to class B. Baseline documentation in accordance with MIL-F-28861 is required and shall be submitted to the qualifying activity prior to completion of quality conformance inspection on the initial lot. Retention of qualification shall be in accordance with class S requirements.

In-process inspection: In accordance with class S requirements, except that the requirements of MIL-C-123 shall not apply.

Quality conformance inspection: The inspection lot definition and inspection of product for delivery shall be in accordance with the class S requirements.

Product assurance level: In accordance with table I.

Part or identifying number: The part or identifying number shall be as follows:



Marking: Filters and capacitors shall be marked, as a minimum, with the following information:

a. Military part or identifying number.

b. JAN brand.

c. Source code.

d. Date code.

e. Terminal identification or circuit diagram (non-symmetrical filters only).

In addition, full marking, in accordance with MIL-F-28861 shall be marked on the package.

Cataloging information. Dash numbers 002, 004, 006, 008, 010, 012, 014, 016, 018, and 020 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 001, 003, 005, 007, 009, 011, 013, 015, 017, and 019 shall be cataloged under FSC 5915 as radio frequency interference filters.

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Ⓐ TABLE I. Electrical characteristics.

Dash number	Product assurance level Class A	Class B	Class S	Style	Circuit	Rated voltage DC volts	Rated voltage AC 1/ volts	Minimum capacitance (µF)	Minimum insertion loss (dB) in accordance with MIL-STD-220 2/ 3/ At +25°C: 30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	At -55°C and +125°C: 30 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz
001, 011	---	X	---	FS10	L2	50	---	1.2	15	28	33	40	40	70	70	13	26	31	38	38	70	70
002, 012	---	X	---	FS10	C	50	---	1.2	15	28	33	40	40	70	70	13	26	31	38	38	70	70
003, 013	X	X	X	FS11	L2	70	---	0.7	10	24	30	40	40	64	70	8	22	28	38	38	64	70
004, 014	X	X	X	FS11	C	70	---	0.7	10	24	30	40	40	64	70	8	22	28	38	38	64	70
005, 015	X	X	X	FS12	L2	100	---	0.45	6	19	25	36	40	60	70	4	17	23	34	38	60	70
006, 016	X	X	X	FS12	C	100	---	0.45	6	19	25	36	40	60	70	4	17	23	34	38	60	70
007, 017	X	X	X	FS13	L2	150	---	0.25	---	14	20	31	40	56	70	---	12	18	29	38	56	70
008, 018	X	X	X	FS13	C	150	---	0.25	---	14	20	31	40	56	70	---	12	18	29	38	56	70
009, 019	X	X	X	FS14	L2	200	125	0.15	---	10	16	26	40	52	70	---	9	14	25	38	52	70
010, 020	X	X	X	FS14	C	200	125	0.15	---	10	16	26	40	52	70	---	9	14	25	38	52	70

1/ 0 to 400 Hz.
2/ Insertion loss measurements shall be made under full load over the frequency range of 150 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no load.
3/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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